

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY



PRACTICAL TRAINING REPORT

REPORT TITLE: VLAN PRINCIPLES AND CONFIGURATION

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**PROGRAM: BACHELOR OF ENGINEERING IN TELECOMMUNICATION
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**COMPANY/ORGANIZATION NAME: MBEYA UNIVERSITY OF SCIENCE AND
TECHNOLOGY**

TRAINING OFFICER'S NAME: MR. FRANK IBRAHIM

UNIVERSITY SUPERVISOR'S NAME: MR. MONTE KAYOKA

PREFACE

I conducted my IPT (Industrial Practical Trainings) at Mbeya University of Science and Technology (MUST) that gave me great chance to observe, learn, experience and practice the basics of Information and communication Technology in real life in short.

I had a chance of doing different tasks that I studied before in class that involved the Network, Cloud, Computing, and Ascend AI Tracks, which examine theoretical knowledge and hands-on ICT skills.

ACKNOWLEDGEMENT;

I would like to extend my grateful thanks to the almighty **God** for his grace, blessings, love and good health every day and during my Practical Training, Also I would like to give a special thanks to my university telecommunication lectures and facilitators (staff), friends and relatives who have been supporting me during all ten weeks of my Practical Training.

I am also grateful to Mbeya University of Science and Technology and all other staff members for accepting my request of conducting the practical training at Mbeya University of Science and Technology and most importantly for orienting me with the real work environment by letting me to be part of their team for the whole period of practical training.

Much thanks also go to the chancellor of MUST allowing me to attend my field practical training at his institution, Also I would like to give a special thanks to my supervisor Mr. Frank Ibrahim for their full cooperation during all weeks of practical training.

ABSTRACT

This field work attachment for the first-year student was conducted at Mbeya University of Science and Technology from 14th July to 19rd September 2025 where different activities were conducted and observed and were done in participatory way. Those activities include virtual local area network (vlan) principles and configuration, and covered key areas in fiber optics and computer networking. During this period, I practiced various networking concepts with emphasis on configuring routers and the implementation of both static and dynamic routing protocols.

The report also revealed the importance of practical training. In reality these Practical trainings are essential because they abide with three dimensions of learning which are cognitive, affective and psychomotor whereby they help one to integrate both of them in field through thinking, recalling, doing and feeling as well experiencing.

Practical Trainings are also advantageous because they shape one's behavior and make him/her learn from other staff members the ways of conduct and how to manage the working environment including the Way of dressing, being organized, respecting, managing time and sometime the way to speak during different Meetings(gatherings) which are conducted to specific institution or organization.

DECLARATION

I **BIRGITHA AMSI BASHAGHE**, am here to declare that the contents of this technical report, are the original copy and also original design of what I went through in all my ten weeks of my Industrial/Field practical training. Also, any of what explained within this report is true information especially when explaining more on among the different activities which were performed for ten weeks during Industrial/Field practical training period.

Name.....

Signature.....

Date.....

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LIST OF ABBREVIATION

LAN	- Local area Network
WAN	- Wide area Network
IP	- Internet protocol
RIP	- Routing Information Protocol
EIGRP	- Enhanced Interior Gateway Routing Protocol
OSPF	-Open Shortest Path First
WAP	-Wireless Access Point
UTP	-Unshielded Twisted Pair
MUST	-Mbeya University of Science and Technology
RJ	-Registered jack
IPT	-Industrial Practical Training
VLAN	-Virtual Local Area Network
MOTD	-Message of The Day
DHCP	-Dynamic Host Configuration Protocol
CLI	-Command Line Interface
DUAL	-Diffusing Update Algorithm
VLSM	-Variable Length Subnet Mask
CIDR	-Classless Inter-Domain Routing

PART I GENERAL PART;

HISTORICAL BACKGROUND OF THE FIRM

In 1986, Mbeya Technical College (MTC) was established for the purpose of training students for the award Full Technician Certificate (FTC) under the Russian Training support. ... The long-term plan was to transform Mbeya technical College into a University of Science and Technology through the University Act No.7 of 2005 and the Charter of Mbeya University of Science and Technology, 2013. Via nation council for technical education (NACTE) establishment order No.9 of and section 9 of the NACTE Act of 1997. MTC was officially launched on 1st September, 1986.

The transformation from MTC to MIST and the MUST was done to fulfill a long –term plan, made during the first phase Government, under the president of United Republic of Tanzania, the late Mwalimu Julius Kambarage Nyerere.

The Universities Act, requires Universities to draft Charters and have also required them to approved as a condition for providing University Education. In compliance with the requirements of the Universities Act, Mbeya University of Science and Technology has been established through part II, section 3(1) of the Mbeya University of Science and Technology Charter, (2013).

LOCATION

The university (MUST) is among the best universities we have in our country Tanzania. It offers many different engineering, science courses as well as some business courses.

Mbeya University of Science and Technology has two campuses. The Main Campus is located in Mbeya City in the Southern Highlands of Tanzania 10 km outside the city center and about 3 Km from the Mbeya – Tunduma Highway. The road to MUST branches from the highway at the Iyunga/Ituta -TAZARA area. The Rukwa Campus of Mbeya University of Science and Technology is located at Kianda along Tunduma Sumbawanga road, about 50 km from Sumbawanga.

A PICTURE OF MUST LOCATION



Figure 1 Showing picture of MUST

VISION;

The vision of Mbeya University of Science and Technology is to become the leading centre of excellence for knowledge, skills and applied education in Science and Technology.

MISSION;

The mission of Mbeya University of Science and Technology is to develop academically, technologically and socially competent students, staff and other stakeholders who will be responsive to the broader needs

And challenges of the society specifically by:

- Facilitating appropriate tuition, practical training and support according to the needs of students and other customers.

GRADUATION AWARDS FOR STUDENTS ORDINARY DIPLOMA PROGRAMMES

- Chancellor's prize: Best Overall Finalist Student
- University Prize: Best Overall Finalist Female Student
- Note that Other Awards are to be set by the University

UNDERGRADUATE PROGRAMMES

- Chancellor's prize: Best Overall Finalist Student
- University Prize: Best Overall Finalist Female Student -Other Award to be set by the university

DEPARTMENTAL PRIZES

Other Prizes from departments may also be given to students based on criteria and award set by the departments.

OTHER PRIZES

Other Prizes from individuals or institutions may also be given to students based on criteria and award set by the donor.

MUST ORGANISATION STRUCTURE AND MANAGEMENT

Management of Mbeya University of Science and Technology (MUST)

Organization structure of Mbeya University of Science and Technology (MUST)

MANAGEMENT OF MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

- Vice Chancellor
Prof. Aloyce Mvuma - PhD (Dortmund) MSc. Eng. (UDSM), B.Eng. (Mysore),
- Acting Deputy Vice Chancellor Planning Finance and Administration (DVC-PFA)
Dr Alexander Mtawa, PhD Mech. Eng. (CPUT, SA), MSc. Mech. Eng. (SOFIA, Bulgaria),
FTC Mech. Eng. (ATC)
- Acting Deputy Vice Chancellor Academic Research and Consultancy (DVC-ARC)
Prof. Osmund Kaunde

- Principal of College of Engineering and Technology
Dr. Zacharia Katambara - PhD (WITS), MSc Eng. (UDSM) BSc Eng. (UDSM)
- Director of Human Resource Management
Mrs. Devotha Msuya - MSc.HRM (MZUMBE)
- Dean of School of Business Studies
Dr. Vicent Kipene – PhD (SUA), MBA(SUA), BSc Ed(SUA), Dip.Ed.(Monduli)
- Principal of College of Science and Education
- Director of Undergraduate Studies
- Dean of students Mr Augustine P. Matem - MA Ed. (UDOM), BSc Ed. (UDSM), Dip Ed. (Mkwawa)
- chief of internal Auditor
Mr. Deusdedit Sinda - CPA (T), Diploma in accountancy (slet), ADA (IFM)
- Chief Accountant Mr. Shwaibu Shaban - CPA (T), PGD Finance Management (IFM), ADA (IFM)
- Chief Planning Officer
Chief Planning Officer Mr. Cornell Msemwa - MBA (SUA), BSc. FST (SUA)
- Director of Library Services
Mr. Novatus Luanda - MA. (UDSM), BA. (UDSM), Dip. (Nyegezi)
- Director of Postgraduate Studies, Research and Publication
- Director Of MUST Consultancy Bureau (MCB)
Mr. Dua Chengula-PhD (Dortmund) MSc. Eng (UDSM) BSc. Civil (UDSM)
- Head-Center for Quality Assurance and Quality Control
Ms. Albentina Leonard, MBA-Finance & Banking (MZUMBE), BA. Ed (UDSM),
- Corporate Counsel
Advocate Saumu Mruke -LLM (Ruaha), PGD in Law (Law School of Tanzania), LLB (MZUMBE)

ORGANIZATION STRUCTURE OF MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

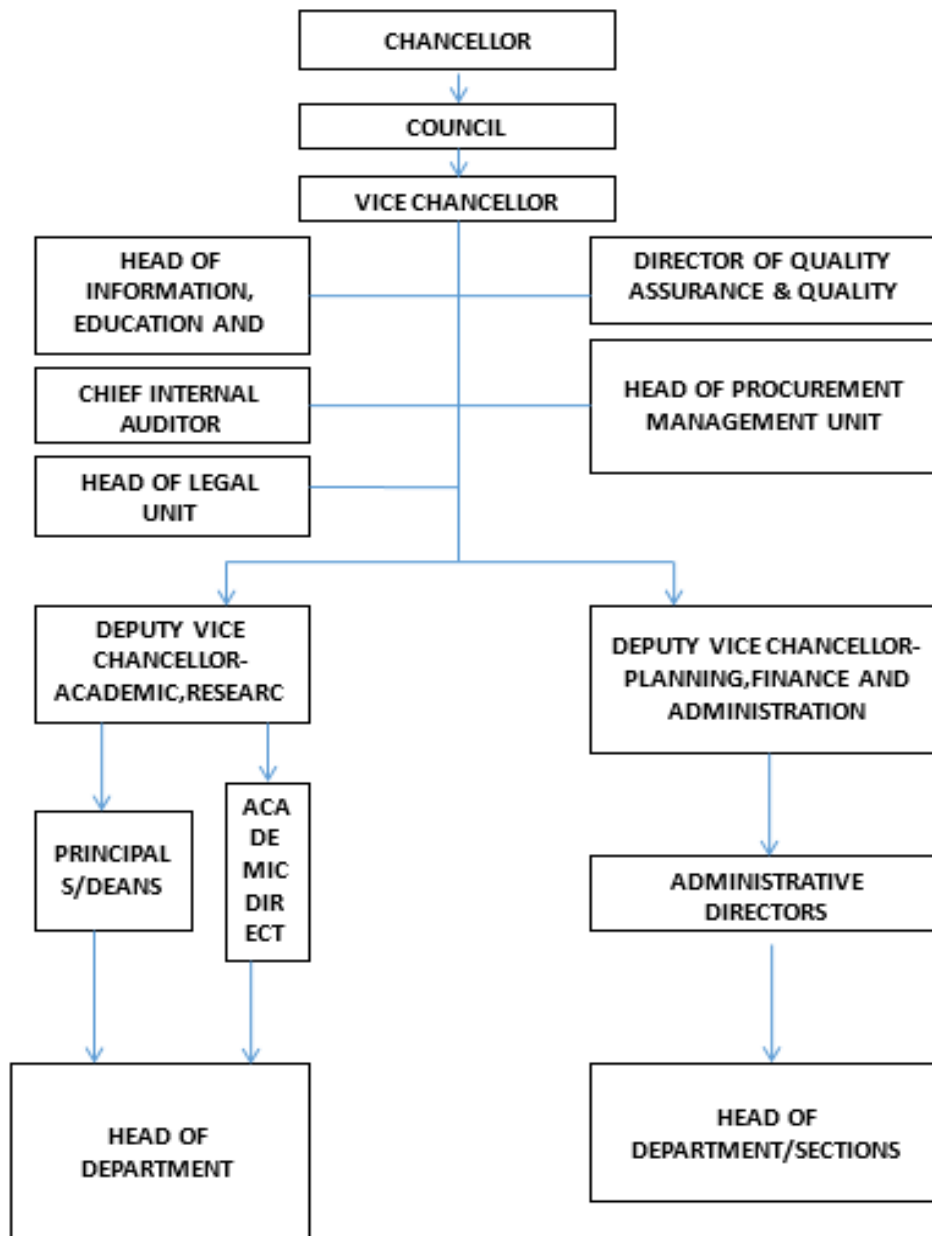


Figure 2 Shows MUST organization structure

PART II TECHNICAL PART:

2.0 Introduction

This chapter provides an overview of the practical activities I was involved in during my Industrial Practical Training (IPT) at Mbeya University of Science and Technology (MUST). The tasks covered various areas of computer networking, server configuration, security tools, and troubleshooting, all of which contributed to building my technical skills and hands-on experience

2.1 Overview of Activities Done During IPT

i) Termination of UTP Cable

I learned how to terminate Unshielded Twisted Pair (UTP) cables using RJ-45 connectors. This involved stripping the cable, arranging the wires in the correct order (T568A or T568B), and crimping the connector using a crimping tool. The cables were then tested for connectivity using a cable tester.

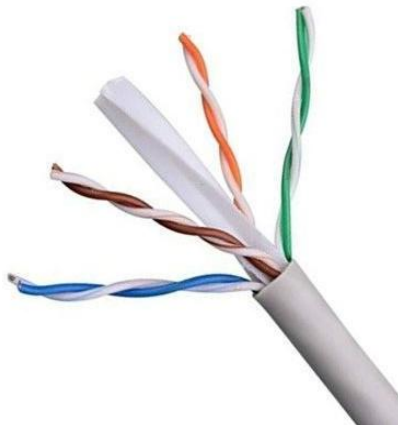


Figure 3 Showing UTP



Figure 4 Showing Ethernet Cable

ii) Switch Configuration

I participated in basic and advanced switch configuration tasks, including assigning IP addresses, configuring ports, and enabling port security to control access to the network.

iii) VLANs and Inter-VLAN Routing

We created and configured VLANs (Virtual LANs) to segment the network into different broadcast domains. This aims at reducing the broadcast domains for effective network performance and security, since only devices in same VLAN can communicate. However, I was introduced to Inter-VLAN routing which was then implemented using a Layer 3 switch or router to enable communication between VLANs.

iv) Router Configuration

Router setup was a key part of my training. This included assigning IP addresses to interfaces, setting up routing protocols, and managing routing tables to ensure proper data packet forwarding.

v) Routing Protocols

I was introduced to different routing protocols such as RIP, OSPF, and EIGRP. I configured and tested them in a simulated and practical environment to understand how routers communicate and update routes dynamically.

vi) Fiber Optic Splicing

Also, we had a session on fiber optic technology, where I observed and practiced basic fiber optic splicing techniques, including cleaving and fusion splicing.

vii) Troubleshooting and Maintenance

we went through the mechanisms and working principles of a printer and a photocopier machine. After, we looked on troubleshooting and maintenance based on photocopier machine. Checked on common issues such as paper jam, toner low, low printing quality, wrong paper size, dirty glass and damaged drum.

vii) WAP Configuration

Wireless Access Points (WAPs) were configured to provide wireless connectivity within the university. I practiced in setting SSIDs, securing wireless networks managing access control



Figure 5 Showing Access Point

viii) Windows Server and Kali Linux Installation

I participated in installing and setting up Windows Server for basic roles such as DHCP and DNS. Additionally, I was introduced to Kali Linux installation and basic penetration testing tools it provides for network security assessments.

2.2 VLAN PRINCIPLES AND CONFIGURATION

2.2.1 INTRODUCTION

A virtual area network (VLAN) is logical segmentation of a physical network into multiple isolated broadcast domains. VLAN allow network administrators to group devices logically, regardless of their physical location, improving security, efficiency and traffic management.

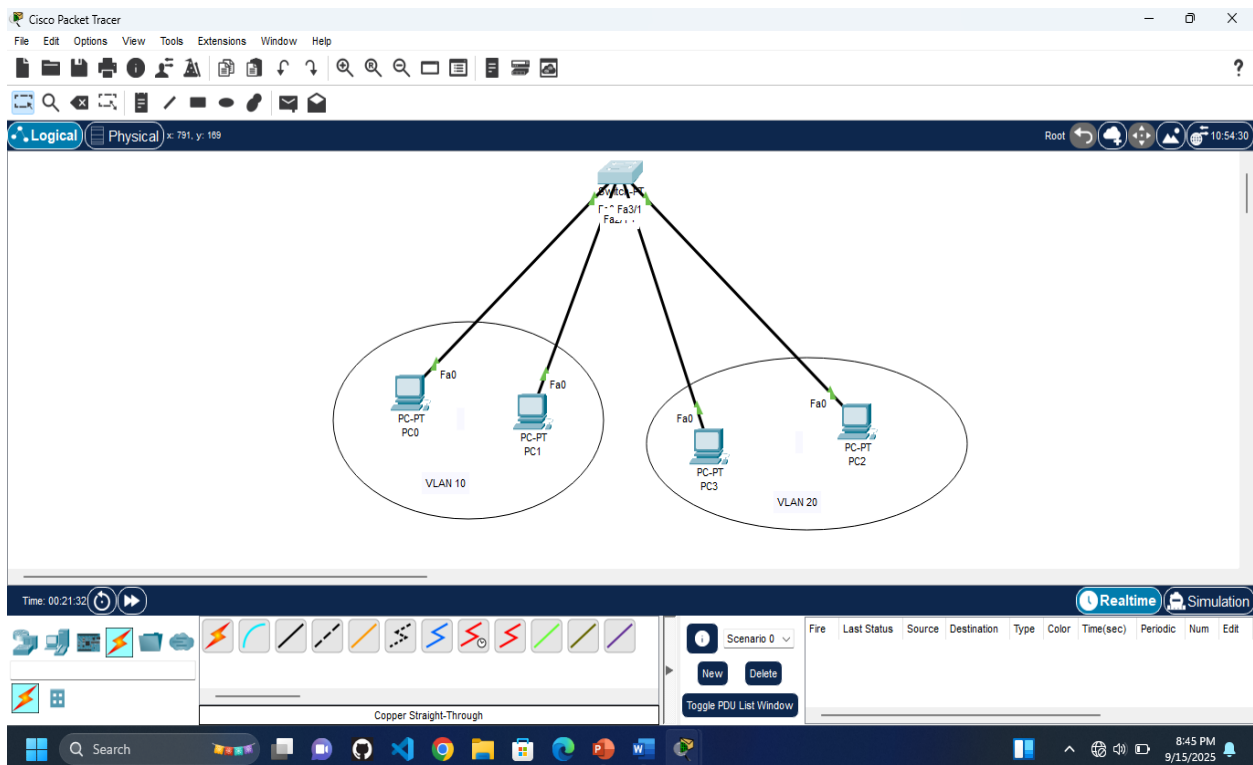


Figure 6 Showing Packet tracer

PROBLEM STATEMENT

Problem statement of VLAN technology is to isolate broadcast domains.

In early, Issues Facing a Traditional Ethernet

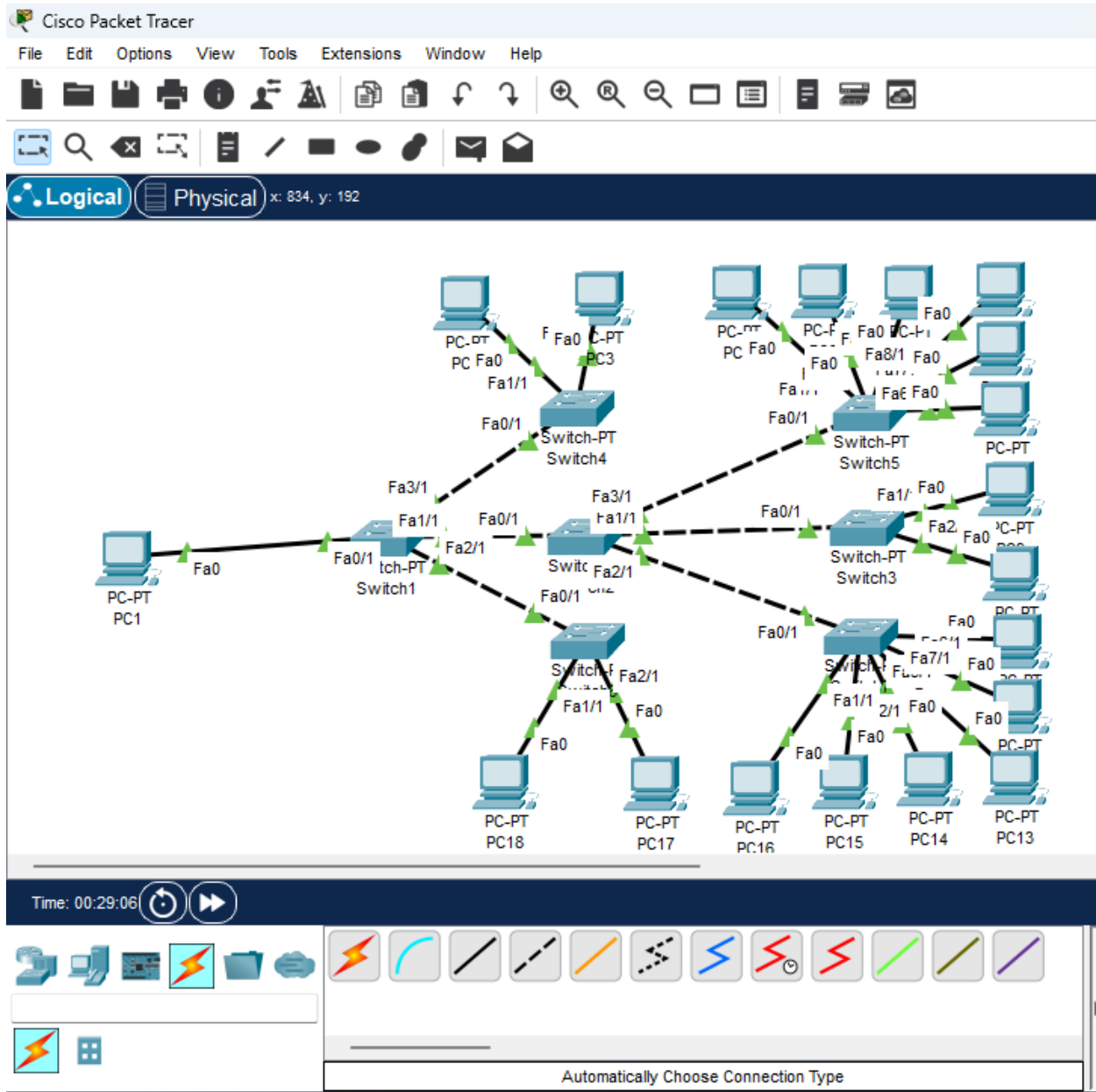


Figure 7 Showing VLAN configuration using packet tracer

- Broadcast domain.

The figure above shows a typical switching network with only PCs and switches. If PC1 sends a broadcast frame, the switches flood the frame on the network. As a result, all the other PCs receive the frame.

The range that broadcast frames can reach is called a Layer 2 broadcast domain (broadcast domain for short). A switching network is a broadcast domain.

- Network security and junk traffic problems.

Assume that PC1 sends a unicast frame to PC2. The MAC address entry of PC2 exists in the MAC address tables of SW1, SW3, and SW7 rather than SW2 and SW5. In this case, SW1 and SW3 forward the frame in point-to-point mode, SW7 discards the frame, and SW2 and SW5 flood the frame. As a result, although PC2 receives the unicast frame, other PCs on the network also receive the frame that should not be received.

The VLAN technology is introduced to solve the problems caused by large broadcast domains. By deploying VLANs on switches, you can logically divide a large broadcast domain into several small broadcast domains. This effectively improves network security, lowers junk traffic, and reduces the number of required network resources. Consider the diagram below

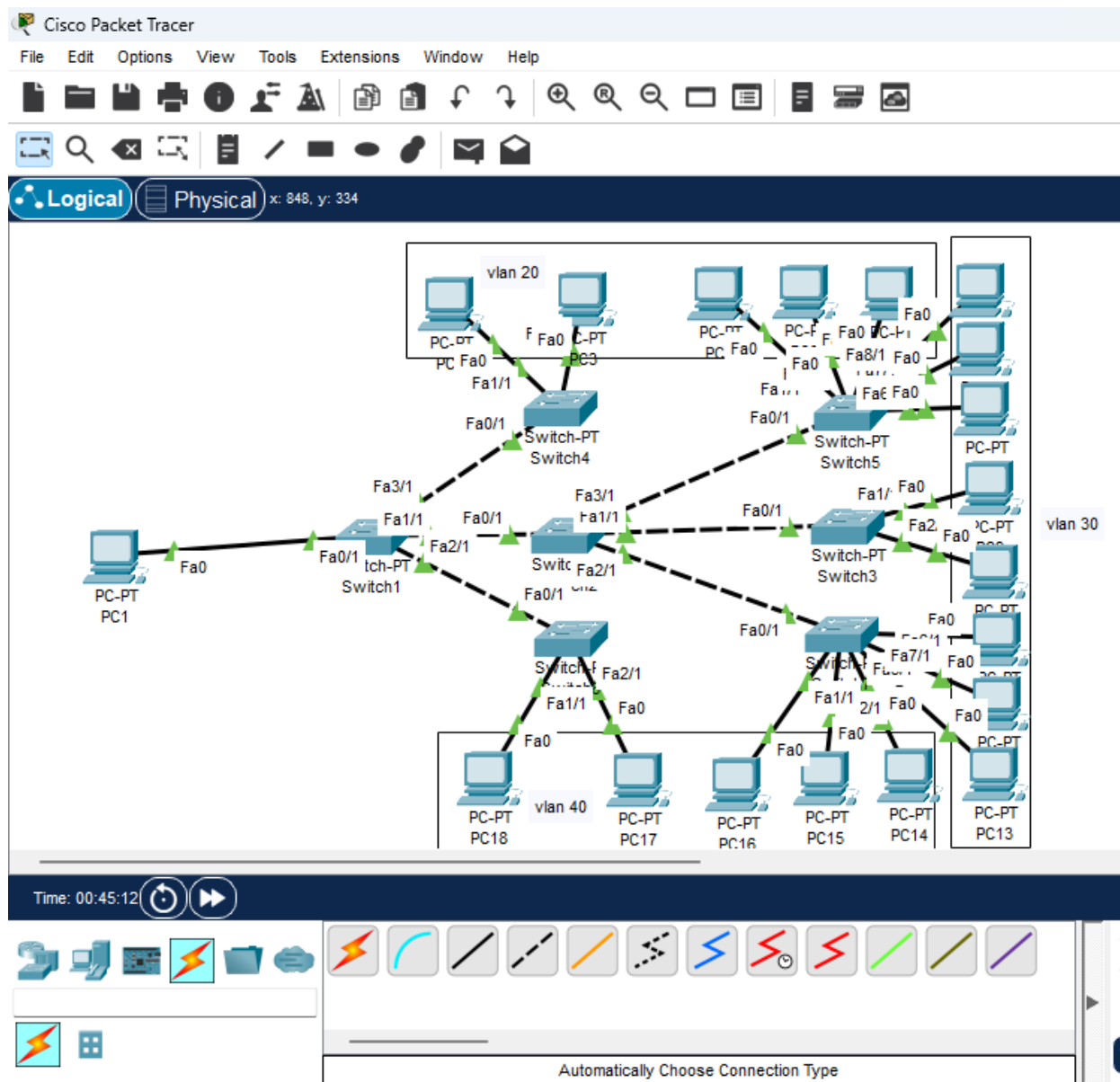


Figure 8 Showing interface configuration using packet tracer

VLAN characteristics.

Each VLAN is a broadcast domain. Therefore, PCs in the same VLAN can directly communicate at Layer 2. PCs in different VLANs, by contrast, can only communicate at Layer 3 instead of directly communicating at Layer 2. In this way, broadcast packets are confined to a VLAN.

Advantages of the VLAN technology.

- i. Allows flexible setup of virtual groups. With the VLAN technology, terminals in different geographical locations can be grouped together, simplifying network construction and maintenance.
- ii. Conserving bandwidth and improving network processing capabilities, by confines each broadcast domain to a single VLAN.
- iii. Enhances LAN security. Frames in different VLANs are separately transmitted, so that PCs in a VLAN cannot directly communicate with those in another VLAN.
- iv. Improves network robustness. Faults in a VLAN do not affect PCs in other VLANs.

2.2.2 VLAN principles

VLAN Implementation

- Switch1 and Switch2 belong to the network of the same enterprise. VLANs are planned for the network, with VLAN 10 for department A and VLAN 20 for department B. Employees in departments A and B are connected to both Switch1 and Switch2.
- Assume that a frame sent from PC1 reaches Switch2 through the link between Switch1 and Switch2. If no processing is implemented, Switch2 can neither identify the VLAN to which the frame belongs nor determine the local VLAN to which the frame should be sent.

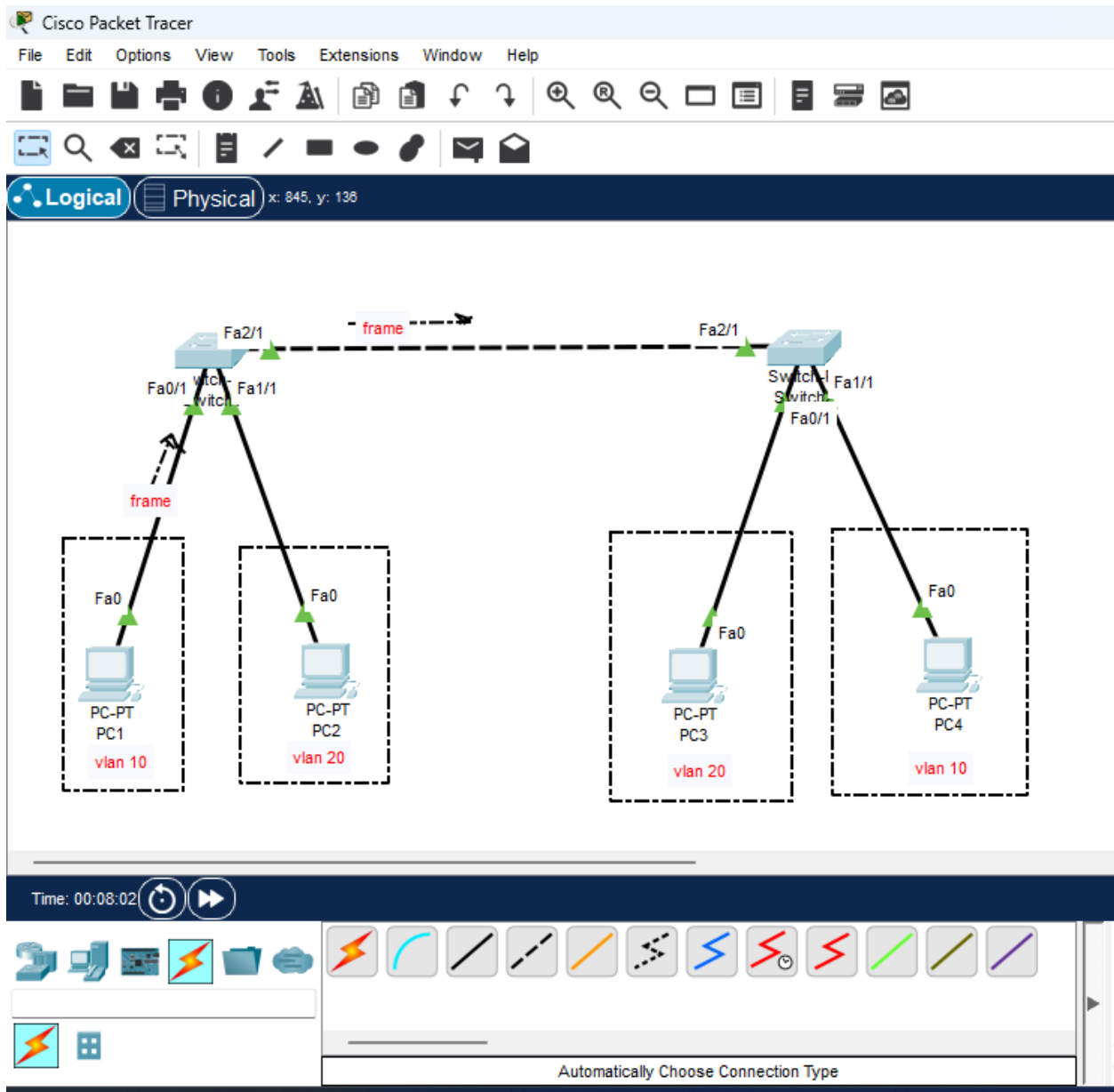


Figure 9 showing sending packets from one router to another

- The link between Switch1 and Switch2 carries data of multiple VLANs. In this situation, a VLAN-based data tagging method is required to distinguish the frames of different VLANs.
- IEEE 802.1Q, often referred to as Dot1q, defines a system of VLAN tagging for Ethernet frames by inserting an 802.1Q tag into the frame header to carry VLAN information.

2.2.3 VLAN Assignment Methods

VLAN can be assigned on network through five ways, that are

- 1) Interface-based assignment: assigns VLANs based on switch interfaces.

A network administrator preconfigures a port VLAN ID (PVID) for each switch interface. When an untagged frame arrives at an interface of a switch, the switch adds a tag carrying the PVID of the interface to the frame. The frame is then transmitted in the specified VLAN.

- 2) MAC address-based assignment: assigns VLANs based on the source MAC addresses of frames.

A network administrator preconfigures the mapping between MAC addresses and VLAN IDs. After receiving an untagged frame, a switch adds the VLAN tag mapping the source MAC address of the frame to the frame. The frame is then transmitted in the specified VLAN.

- 3) IP subnet-based assignment: assigns VLANs based on the source IP addresses and subnet masks of frames.

A network administrator preconfigures the mapping between IP addresses and VLAN IDs. After receiving an untagged frame, a switch adds the VLAN tag mapping the source IP address of the frame to the frame. The frame is then transmitted in the specified VLAN.

- 4) Protocol-based assignment: assigns VLANs based on the protocol (suite) types and encapsulation formats of frames.

A network administrator preconfigures the mapping between protocol (suite) types and VLAN IDs. After receiving an untagged frame, a switch adds the VLAN tag mapping the protocol (suite) type of the frame to the frame. The frame is then transmitted in the specified VLAN.

- 5) Policy-based assignment: assigns VLANs based on a specified policy, such as a policy combining the preceding methods.

A network administrator preconfigures a policy. After receiving an untagged frame that matches the policy, a switch adds a specified VLAN tag to the frame. The frame is then transmitted in the specified VLAN.

2.2.4 SWITCH INTERFACE

Switch interface types are,

I. Access interface

An access interface often connects to a terminal (such as a PC or server) that cannot identify VLAN tags, or is used when VLANs do not need to be differentiated.

II. Trunk interface

A trunk interface often connects to a switch, router, AP, or voice terminal that can receive and send both tagged and untagged frames.

2.2.5 VLAN PLANNING

- **VLAN assignment rules**

VLAN IDs can be randomly assigned within the supported range. To improve VLAN ID continuity, you can associate VLAN IDs with subnets during VLAN assignment.

- By service: voice, video, and data VLANs
- By department: e.g. VLANs for engineering, marketing, and financing departments
- By application: e.g. VLANs for servers, offices, and classrooms

- **Example for VLAN planning**

- Assume that there are three buildings: administrative building with offices, classrooms, and financing sections, teaching building with offices and classrooms, and office building with offices and financing sections. Each building has one access switch, and the core switch is deployed in the administrative building.

➤ The following table describes the VLAN plan.

VLAN ID	IP Address Segment	Description
1	X.16.10.0/24	VLAN to which office users belong
2	X.16.20.0/24	VLAN to which the users of the financing department belong
3	X.16.30.0/24	VLAN to which classroom users belong
100	Y.16.100.0/24	VLAN to which the device management function belongs

2.2.6 Basic VLAN Configuration Commands

This is achieved by creating a VLAN on the switch's CLI, assigning ports to that VLAN using switchport mode access and switchport access vlan commands, and establishing trunk ports to carry multiple VLANs using switchport mode trunk. The process enables network administrators to improve security by isolating broadcast domains and manage network traffic more effectively.

2.2.7 Steps to Configure VLANs in Packet Tracer

1. Access the Switch's CLI: Connect a PC to the switch using a console cable and launch the Terminal to access the switch's command-line interface (CLI).
2. Enter Global Configuration Mode: Enable privileged EXEC mode using enable and then enter global configuration mode with configure terminal.
3. Create VLANs: Create new VLANs using the vlan <vlan_id> command, then assign a name using the name <vlan_name> command for clarity.

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Sales
```

4. Assign Access Ports: Assign switch ports to a specific VLAN by configuring them as access ports and specifying the desired VLAN.

```
Switch(config)# interface <interface_id> (e.g., interface fastethernet 0/1)
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 10
```


5. Configure Trunk Ports: On links connecting switches or a switch to a router, use the switchport mode trunk command to allow multiple VLANs to pass through.

Switch(config)#

interface <interface_id>

(e.g., interface fastethernet 0/24)

Switch(config-if)# switchport mode trunk

You can also restrict which VLANs are allowed on the trunk with switchport trunk allowed vlan <vlan_ids>.

6. Verify the Configuration: Use the show vlan brief command in the privileged EXEC mode to verify the VLAN and port assignments.

2.2.8 Application and Benefits in Packet Tracer

- i. Network Segmentation: VLANs divide a large network into smaller, isolated logical segments, similar to having multiple physical LANs.
- ii. Enhanced Security: By creating separate VLANs, broadcast traffic is limited to within the respective VLAN, preventing devices in one VLAN from receiving traffic from another unless specifically routed.
- iii. Improved Performance: Limiting broadcasts reduces network congestion, leading to better performance for devices within their assigned VLANs.
- iv. Flexibility and Scalability: VLANs make it easier to manage network resources and add or remove devices without having to change the physical cabling infrastructure.

PART III CONCLUSSION AND RECOMMENDATIONS

3.1 RECOMMENDATIONS

Based on the training I received, I would recommend that future sessions continue to focus on practical exposure, especially in configuring and troubleshooting real network devices. While theory is important, I believe working directly with routers, switches, and simulation tools helped me understand better.

I would like to advice the head of Mbeya university of science and Technology (MUST) that more time be allocated for practical sessions and advanced topics such as network security configurations, access control lists (ACLs), and complex routing scenarios. These areas are essential in preparing students for actual job environments.

Lastly, I encourage students undergoing similar training to fully participate with the tasks, ask questions, and collaborate with others as this made my learning experience more effective and enjoyable during IPT.

Conclusion

The Industrial Practical Training (IPT) period at the [Company Name] Network Department has been an invaluable and transformative experience, successfully bridging the gap between academic theory and the practical demands of a live network environment. Over the course of the training, I was actively involved in key operational areas, including network monitoring and troubleshooting, assisting with the configuration of switches and routers, managing physical cable infrastructure, and contributing to network documentation.

This hands-on exposure was instrumental in deepening my understanding of core networking principles such as IP subnetting, VLAN segmentation, and network security protocols. I gained significant practical proficiency with essential tools and commands, moving beyond theoretical concepts to solve real-world problems. Furthermore, the experience highlighted the critical importance of systematic procedures, clear documentation, and effective teamwork in ensuring network stability and security.

I would like to express my sincere gratitude to the management of [Company Name] and my supervisor, [Supervisor's Name], for their invaluable guidance, patience, and for providing me with this excellent learning opportunity. The knowledge and skills acquired during this training

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